



Figure 6: OpenVZ virtual machine creation made easy

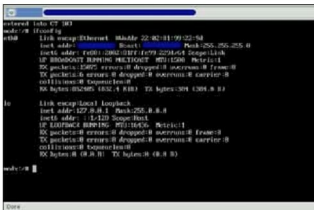


Figure 7: VM console access via Java VNC applet

```
# vgcreate -v VG /dev/cciss/c0d0p3
```

Success! I have allocated 150 GB for a logical volume where OpenVZ containers will reside:

```
# lvcreate -L 150G -n VG_150Gb -v VG
```

Format the logical volume, for example, as an `ext4` filesystem:

```
# mkfs.ext4 -m 1 -l OpenVZ_Space -v /dev/mapper/VG-VG_150Gb
```

Add a new entry into `/etc/fstab`:

```
/dev/mapper/VG-VG_150Gb    /var/lib/vz    ext4
errors=remount-ro    0    1
```

Now, this new volume will be mounted automatically after each reboot, at `/var/lib/vz`. Let's analyse this. The `'images'` subdirectory contains file images of QEMU containers; `'root'` and `'private'` subdirectories are reserved for OpenVZ containers. The directory `'template/cache'` is where you should place virtual appliance archives, and the `'template/ISO'` directory is for CD/DVD installation images.

```
$ ls /var/lib/vz
backup dump images lock lost-found private root template
vztmp
```

We still have 173 GB of free LVM space that we should reserve exclusively for QEMU containers:

```
# vgdisplay
VG Size      323.84 GB
PE Size      4.00 MB
Total PE     82697
Alloc PE / Size 38400 / 150.00 GB
Free PE / Size 44297 / 173.84 GB
```

OpenVZ VMs

As mentioned earlier, Proxmox uses two types of virtualisation—OpenVZ and QEMU. The first is para-virtualisation, where the processes, filesystem and memory of a virtual container actually reside on a host machine, but are logically 'isolated' by OpenVZ technology. Due to architectural constraints of OpenVZ, only Linux distributions can run as para-virtualised machines.

The Proxmox team distributes OS Templates and 'Virtual Appliances' (packed OpenVZ containers) on their site. You can download these and upload them to your infrastructure via 'VM Manager > Appliance Templates > Upload File'. These are preconfigured and ready-to-use Debian, Ubuntu, CentOS and Fedora distributions, with a minimum set of packages pre-installed. Several appliances are completely preconfigured versions of SugarCRM, MediaWiki, Drupal, etc. All you need is to point to 'Virtual Machines > Create', choose the type of VM as 'Container (OpenVZ)', and select the template (Figure 6)... that's it!

Alternatively, you can create an OpenVZ machine from the command line, as follows:

```
# /usr/bin/pvecl vzcreate 101 -disk 8 \
--ostemplate local:vztmpl/debian-5.0-standard_5.0-2_amd64.tar.gz \
--rootpasswd $1$PwC1Mv$KX5dbbyAmv03nt15C2f0. \
--hostname wiki.local --nameserver 10.10.1.200 --searchdomain local \
--onboot yes --swap 1024 --mem 1024 --cpus 1 \
--netif ifname=eth0,mac=66:2E:57:66:2E:36,bridge=vzbr5
```

Here, the first four parameters mean the following:

- Create a VM with ID 101
 - Allocate disk space of 8 GB
 - Use the `debian-5.0-standard_5.0-2_amd64.tar.gz` template
 - Supply a `cryptd` (by the `glibc2 crypt()` function) password for the root account of this VM.
- Other parameters are self-explanatory. You can omit the